

Standard Operating Procedure (SOP)

Title

SOP for Creating New and Updating Existing QC Metric Definitions for GA4GH WGS-QC Standards v2.0

Version

- SOP Version: 1.0
 - Maintained by: GA4GH WGS-QC Workgroup
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1. Purpose

The purpose of this SOP is to define a clear, standardized process for GA4GH WGS-QC workgroup members and subject-matter experts to contribute to the development of **WGS-QC Standards v2.0**. This includes:

- Creating **new QC metric definitions** for short-read and long-read WGS
- Updating or amending **existing QC metric definitions from v1.0**
- Differentiating QC metrics by **sequencing data type (short-read vs long-read)**

This SOP supports the **roadmap v2** activities and ensures consistency, traceability, and reviewability of QC metric definitions prior to publication.

2. Scope

This SOP applies to: All contributors to the WGS-QC Standards GitHub repository

- Development of QC metrics for:
 - Short-read WGS
 - Long-read WGS (ONT, PacBio, Constellation)
- Metric definition work contributing to **WGS-QC Standards v2.0**

Out of scope:

- Tool benchmarking or performance evaluation / Dataset-specific QC thresholds
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3. References

- GA4GH WGS-QC GitHub Repository (v1.0):
<https://github.com/ga4gh/quality-control-wgs>
 - Metric Definition Template (current):
https://ga4gh.github.io/quality-control-wgs/metrics_definitions/
 - Published specifications (gh-pages):
<https://github.com/ga4gh/quality-control-wgs/tree/gh-pages>
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4. Roles and Responsibilities

4.1 Contributors

- Propose new QC metrics or updates to existing metrics
- Author metric definitions using the agreed template
- Submit changes via GitHub pull requests

4.2 Reviewers (WGS-QC Workgroup)

- Review metric definitions for clarity, correctness, and alignment with consensus
- Provide feedback and request revisions where required
- Approve and merge contributions

4.3 Maintainers

- Manage branches and releases
 - Publish approved metric definitions to `gh-pages`
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5. Overview of Contribution Workflow

1. Clone the GA4GH WGS-QC repository
 2. Create a new feature branch
 3. Create or update QC metric definition files (`.md`)
 4. Commit and push changes
 5. Create a pull request (PR)
 6. Review, revise, and merge into `develop-definition-v2`
 7. Publish approved metrics to `gh-pages`
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6. Metric Definition Requirements (v2.0)

Each QC metric definition **must** follow the latest workgroup-consensus template and be authored as a **Markdown (.md)** file.

6.1 Mandatory and Optional Fields

Each metric definition must include the following sections:

- **ID (mandatory):** Unique metric identifier (e.g., `yield_bp_q30`)
- **Description (mandatory):** Clear description of what the metric measures
- **Implementation details (mandatory):** How the metric is calculated or derived
- **Type (mandatory):** Whether the metric value is an integer or float. For float values, the minimal required decimal precision must be defined to ensure consistent reporting across workflows.
- **Functionally equivalent implementation (optional):** Alternative implementations/tools
- **Data type (mandatory):** `short-read` | `long-read`
- **Assembly (mandatory):** `GRCh37` | `GRCh38`
- **Version (mandatory):** `1.0` | `2.0`
- **Sequencing platform / Technology (optional):** `Illumina` | `Constellation` | `ONT` | `PacBio`
- **Associated aligner (optional):** Free text or controlled list (e.g., `BWA-MEM`, `DRAGEN`, `Minimap2`..)
- **Associated variant caller (optional):** Free text or controlled list (e.g., `GATK`, `DRAGEN`, `DeepVariant`..)

6.2 Versioning Guidance

- Existing v1.0 metrics that are **amended or extended** for v2.0 must:
 - Clearly state `Version: 2.0`
 - Document changes or clarifications in the implementation details
- New metrics introduced in Roadmap v2 must be labelled as `Version: 2.0`

7. GitHub Contribution Procedure

7.1 Clone Repository

```
git clone https://github.com/ga4gh/quality-control-wgs.git
cd quality-control-wgs
```

7.2 Create Feature Branch

Create a new branch for your contribution:

```
git checkout -b feature/<short-description>
```

Example:

```
git checkout -b feature/long-read-coverage-metric
```

7.3 Navigate to Metric Definitions Directory

```
cd docs/metrics_definitions
```

7.4 Create or Update Metric Definition

- Create a new `.md` file using the metric definition template, **or**
- Amend/update an existing metric definition to:
 - Correct errors
 - Improve clarity
 - Differentiate short-read vs long-read sequencing

File naming should be descriptive and consistent with existing metrics.

8. Commit and Push Changes

Stage and commit your changes:

```
git add <file(s)>
git commit -m "Add/update QC metric definition for <metric name>"
```

Push your feature branch to origin:

```
git push origin feature/<short-description>
```

9. Pull Request (PR) Process

1. Create a Pull Request (PR) from your feature branch
2. Target branch: **develop-definition-v2**
3. Provide a clear PR description including:
 - Purpose of the metric
 - Whether it is new or an update to v1.0
 - Any sequencing-type-specific considerations
 - Links to or references for any relevant GitHub issues, pull requests, or discussion threads

9.1 Recommended Pull Request (PR) Description Template

```
## Summary
Brief description of the QC metric being added or updated.

## Metric Scope
- Metric name / ID:
- New metric or update to existing v1.0 metric:
- Data type: short-read | long-read

## Rationale
Explain why this metric is required or why changes are proposed.

## Implementation Notes
Key implementation details, assumptions, or sequencing-technology-specific
considerations.

## Related Discussions
- GitHub issue(s): <link>
- Pull request(s): <link>
- Workgroup discussion / meeting notes (if applicable): <link>
```

Contributors are encouraged to use the following template when creating a pull request:

- PRs will be reviewed by WGS-QC workgroup members
- Contributors may be asked to revise content based on feedback
- Once approved, PRs will be merged into `develop-definition-v2` branch

10. Publication

- Approved metric definitions will be published via the `gh-pages` branch
- Published content represents the **official WGS-QC v2.0 product specification**

11. Change Management

- All changes must be tracked via GitHub commits and PRs
 - Major updates or breaking changes should be discussed during workgroup meetings
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12. Contact

For questions or clarification, please contact the GA4GH WGS-QC workgroup via the GitHub repository or scheduled workgroup meetings.

List of Potential Long-read QC Metrics

The following non-exhaustive list represents **candidate QC metrics** for consideration as part of the WGS-QC v2.0 roadmap. These metrics are provided to guide discussion, prioritization, and contribution by the workgroup and domain experts.

Read- and Coverage-level Metrics

Read N50
Yield (bp)
Yield (reads $\geq$ N bp): 0, 10,000, 50,000, 100,000
Read number ?
Reads mapped
Reads mapped (%)
Median read quality / Mean read quality
Median read length / Mean read length
Autosomal depth (mean), non-gap white listed regions
Bases with $\geq$ N-fold coverage (e.g. 5X / 10X / 15X)
Genome coverage uniformity
Contamination

Variant-level Metrics

SNV count
Indel count (short indels | long indels)
Heterozygous / Homozygous ratio (SNVs | Indels)
Transition / Transversion (Ti/Tv) rate
